



Data Dictionary

This data dictionary presents the relational database structure shown in the CrayCare Entity-Relationship Diagram (ERD). Each table lists its exact table name and column names, data type, description, and constraints or default values.

The table and column names are written exactly as defined in the ERD. Primary keys identify each record, while foreign keys connect related records such as users, tanks, batches, sensor data, feeding activities, and notifications.

Table: users

Field Name	Data Type	Description	Constraints / Default
uid	varchar	Firebase Auth UID.	Primary Key
full_name	varchar	Complete name of the user.	Optional
email	varchar	Email address used for sign-in and account recovery.	Unique
role	varchar	owner admin.	Optional
status	varchar	active disabled.	Optional
tank_id	varchar	owned tank (null for admin).	Foreign tanks.tank_id Key:
photo_url	text	Link to the user's profile image.	Optional
created_at	timestamp	Date and time when the record was created.	Optional

Table: notification_settings

Field Name	Data Type	Description	Constraints / Default
uid	varchar	one row per user.	Primary Key; Foreign Key: users.uid
sound	boolean	Stores the sound.	Default: true
vibration	boolean	Stores the vibration.	Default: true
critical	boolean	Stores the critical.	Default: true
warning	boolean	Stores the warning.	Default: true
feeding	boolean	Stores the feeding.	Default: true
sampling	boolean	Stores the sampling.	Default: true
operational	boolean	Push preference for automatic actuator and sensor-recovery updates.	Default: true
updated_at	timestamp	Date and time when the record was last updated.	Optional



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Table: hardware_assignments

Field Name	Data Type	Description	Constraints / Default
id	varchar	always "currentOwner" — single Firestore doc.	Primary Key
owner_uid	varchar	Unique identifier of the tank owner.	Foreign Key: users.uid
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
assigned_by	varchar	admin who assigned.	Foreign Key: users.uid
assigned_at	timestamp	Date and time when the hardware was assigned.	Optional

Table: tanks

Field Name	Data Type	Description	Constraints / Default
tank_id	varchar	== owner uid by default (1 user = 1 tank).	Primary Key
owner_uid	varchar	Unique identifier of the tank owner.	Foreign Key: users.uid
current_batch_id	varchar	nullable.	Foreign Key: batches.batch_id
stocking_date	timestamp	Date when the production batch was stocked.	Optional
last_sample_date	timestamp	Stores the last sample date.	Optional
sample_count	int	Number of valid readings included for sample.	Optional
initial_population	int	Stores the initial population.	Optional
initial_total_sample_weight	decimal(10,2)	Stores the initial total sample weight.	Optional
initial_total_sample_length	decimal(10,2)	Stores the initial total sample length.	Optional
is_initialized	boolean	Shows whether initialized is true or false.	Default: false
created_at	timestamp	Date and time when the record was created.	Optional

Table: sensor_thresholds

Field Name	Data Type	Description	Constraints / Default
id	varchar	sensor_name (temperature ph_level dissolved_oxygen turbidity water_level feed_level).	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
sensor_name	varchar	temperature ph_level dissolved_oxygen turbidity water_level feed_level.	Optional
min_value	decimal(10,2)	Lowest value of the configured normal range.	Optional
max_value	decimal(10,2)	Highest value of the configured normal range.	Optional
critical_value	decimal(10,2)	Feed level only: critical-below percentage.	Optional
hopper_capacity_g rams	decimal(10,2)	Feed level only: calibrated full capacity.	Optional



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Field Name	Data Type	Description	Constraints / Default
updated_at	timestamp	Date and time when the record was last updated.	Optional

Table: sensor_readings

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign tanks.tank_id Key:
temperature	decimal(6,2)	Current water temperature reading.	Optional
ph_level	decimal(5,2)	Current pH reading of the water.	Optional
dissolved_oxygen	decimal(5,2)	Current dissolved oxygen reading.	Optional
turbidity	decimal(8,2)	Current water turbidity reading.	Optional
water_level	decimal(6,2)	Current pond water-level reading.	Optional
feed_level	decimal(5,2)	Hopper percentage; operational, not a WQA feature.	Optional
estimated_feed_grams	decimal(10,2)	Estimate calculated from level percentage and configured hopper capacity.	Optional
recorded_at	timestamp	Date and time when the sensor data was recorded.	Optional

Table: sensor_readings_history

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign tanks.tank_id Key:
temp_min	decimal(6,2)	Lowest recorded temp value.	Optional
temp_max	decimal(6,2)	Highest recorded temp value.	Optional
temp_avg	decimal(6,2)	Average recorded temp value.	Optional
pH_min	decimal(5,2)	Lowest recorded pH value.	Optional
pH_max	decimal(5,2)	Highest recorded pH value.	Optional
pH_avg	decimal(5,2)	Average recorded pH value.	Optional
DO_min	decimal(5,2)	Lowest recorded dissolved oxygen value.	Optional
DO_max	decimal(5,2)	Highest recorded dissolved oxygen value.	Optional
DO_avg	decimal(5,2)	Average recorded dissolved oxygen value.	Optional
turbidity_min	decimal(8,2)	Lowest recorded turbidity value.	Optional
turbidity_max	decimal(8,2)	Highest recorded turbidity value.	Optional
turbidity_avg	decimal(8,2)	Average recorded turbidity value.	Optional
waterLevel_min	decimal(6,2)	Lowest recorded water level value.	Optional
waterLevel_max	decimal(6,2)	Highest recorded water level value.	Optional



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Field Name	Data Type	Description	Constraints / Default
waterLevel_avg	decimal(6,2)	Average recorded water level value.	Optional
feed_level	decimal(5,2)	Operational snapshot; excluded from WQA inputs.	Optional
estimated_feed_grams	decimal(10,2)	Estimated amount of feed remaining in the hopper.	Optional
recorded_at	timestamp	Date and time when the sensor data was recorded.	Optional

Table: actuators

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
device_id	varchar	pump aerator1 aerator2.	Optional
control_mode	varchar	on off auto (requested by owner).	Optional
current_state	varchar	on off (actual relay state from ESP32).	Optional
last_changed	timestamp	Stores the last changed.	Optional

Table: actuator_logs

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
actuator_type	varchar	pump aerator1 aerator2.	Optional
action	varchar	Description of the action that occurred.	Optional
type	varchar	on off auto.	Optional
logged_at	timestamp	Date and time when the event was recorded in the log.	Optional

Table: feeder_status

Field Name	Data Type	Description	Constraints / Default
tank_id	varchar	Unique identifier of the related tank.	Primary Key; Foreign Key: tanks.tank_id
command_id	varchar	Optional manual request correlation; empty for scheduled feeds.	Optional
status_reason	varchar	Human-readable block reason.	Optional
status	varchar	idle checking_feed_level dispensing completed skipped_insufficient blocked.	Optional
dispense_count	int	completed feeds (Firestore: dispenseCount).	Optional



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Field Name	Data Type	Description	Constraints / Default
last_seen	timestamp	Firestore: lastSeen.	Optional
last_dispensed_at	timestamp	Stores the last dispensed at.	Optional
last_dispensed_grams	decimal(8,2)	Stores the last dispensed grams.	Optional
feed_level	decimal(5,2)	Estimated percentage of feed remaining in the hopper.	Optional
estimated_feed_grams	decimal(10,2)	Estimated amount of feed remaining in the hopper.	Optional

Table: feeder_schedules

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
time	varchar	HH:MM.	Optional
ampm	varchar	AM PM.	Optional
time_value	int	minutes since midnight.	Optional
grams	decimal(8,2)	Optional; default 20 g. Supported: 20–200 g, multiples of 20. Servo-cycle estimate, not measured weight.	Optional
days	varchar	day-of-week mask "1111111" (Sunday first, 1=active).	Optional
enabled	boolean	Stores the enabled.	Default: true
is_done	boolean	Shows whether done is true or false.	Default: false
effective_at_ms	bigint	UTC time stored as milliseconds; creation/edit/re-enable instant.	Optional
last_outcome	varchar	completed skipped_insufficient blocked failed; set from device outcome log.	Optional
last_occurrence_at	bigint	UTC time stored as milliseconds of the scheduled minute; outcomes are date scoped, no midnight reset.	Optional
created_at	timestamp	Date and time when the record was created.	Optional

Table: feeder_logs

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
command_id	varchar	Optional originating manual command id; commands are deleted after durable intent.	Optional
action	varchar	Description of the action that occurred.	Optional



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Field Name	Data Type	Description	Constraints / Default
type	varchar	auto manual missed error.	Optional
status	varchar	completed skipped_insufficient blocked failed.	Optional
schedule_key	varchar	Optional originating schedule document id; not a required FK because schedules may be deleted.	Optional
schedule_time	varchar	Optional originating schedule time, e.g. 4:00 PM.	Optional
occurrence_at	bigint	Original UTC milliseconds since January 1, 1970, retained through offline retries.	Optional
requested_grams	decimal(8,2)	Stores the requested grams.	Optional
estimated_dispensed_grams	decimal(8,2)	Completed cycles only; not scale-measured mass.	Optional
amount_basis	varchar	servo_cycle_estimate.	Optional
estimated_available_grams	decimal(10,2)	Stores the estimated available grams.	Optional
feed_level_before	decimal(5,2)	Stores the feed level before.	Optional
feed_level_after	decimal(5,2)	Stores the feed level after.	Optional
level_change_detected	boolean	Supporting indication only; not proof of exact dispensed mass.	Optional
verification_note	varchar	Stores the verification note.	Optional
logged_at	timestamp	Date and time when the event was recorded in the log.	Optional

Table: feeder_commands

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
command_type	varchar	feed_now (manual feeding command).	Optional
grams	decimal(8,2)	optional; 20.0 default when absent.	Optional
issued_by	varchar	Stores the issued by.	Foreign Key: users.uid
issued_at	timestamp	Date and time when the command was created.	Optional
expires_at	timestamp	App request deadline; ESP also enforces at most 60 seconds after issued_at.	Optional

Table: feeder_notification_receipts

Field Name	Data Type	Description	Constraints / Default
log_id	varchar	tanks/{tankId}/feeder_notification_receipts/{logId}.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
uid	varchar	Unique identifier of the user account.	Foreign Key: users.uid



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Field Name	Data Type	Description	Constraints / Default
push_attempt_claimed_at	timestamp	Server-only at-most-once claim, not proof of FCM delivery.	Optional

Table: sensor_ingestion

Field Name	Data Type	Description	Constraints / Default
id	varchar	Logical staging: sensorIngestion/current or current/history/{docId}.	Primary Key
hardware_id	varchar	Firestore field: hardwareId.	Optional
source_tank_id	varchar	Stores the source tank id.	Optional
source_owner_uid	varchar	Stores the source owner uid.	Optional
source_assignment_at_ms	bigint	Capture-time assignment identity, not upload-time ownership.	Optional
captured_at_ms	bigint	Stores the captured at ms.	Optional
routing_status	varchar	quarantined when history cannot be safely assigned.	Optional
routing_reason	varchar	Stores the routing reason.	Optional

Table: machine_learning_assessments

Field Name	Data Type	Description	Constraints / Default
id	varchar	sortable UTC timestamp doc id.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
level	varchar	Good Moderate Poor Critical Insufficient.	Optional
model_level	varchar	Raw ML class: Good Moderate Poor Critical.	Optional
rule_level	varchar	Deterministic rolling-risk safety floor.	Optional
safety_override	boolean	Stores the safety override.	Default: false
source	varchar	Model version Rule-based fallback Insufficient data Stale sensor history.	Optional
model_algorithm	varchar	RandomForestClassifier Rule-based Not applied.	Optional
model_version	varchar	Version embedded in the deployed model artifact.	Optional
training_data_origin	varchar	synthetic during prototype; field when retrained from actual pond history.	Optional
label_origin	varchar	deterministic_threshold_assessment until independent expert labels are available.	Optional
model_feature_count	int	Number of valid readings included for model feature.	Optional
analysis_window_minutes	int	Stores the analysis window minutes.	Default: 60
data_status	varchar	ready insufficient stale.	Optional



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Field Name	Data Type	Description	Constraints / Default
source_recorded_at	timestamp	Latest actual source reading; ISO-8601 in Firestore, nullable.	Optional
source_age_seconds	int	Stores the source age seconds.	Optional
confidence	int	Stores the confidence.	Optional
driver	varchar	DO pH temp turbidity waterLevel overall.	Optional
driver_label	varchar	Stores the driver label.	Optional
driver_value	decimal(10,2)	Stores the driver value.	Optional
driver_unit	varchar	Stores the driver unit.	Optional
driver_min	decimal(10,2)	Lowest recorded driver value.	Optional
driver_max	decimal(10,2)	Highest recorded driver value.	Optional
problem	varchar	Stores the problem.	Optional
insight	text	Stores the insight.	Optional
action	text	Description of the action that occurred.	Optional
ts_epoch	bigint	Stores the ts epoch.	Optional
timestamp	timestamp	Stores the timestamp.	Optional

Table: batches

Field Name	Data Type	Description	Constraints / Default
batch_id	varchar	Unique identifier of the related production batch.	Primary Key
tank_id	varchar	Unique identifier of the related tank.	Foreign Key: tanks.tank_id
batch_status	varchar	active harvested superseded.	Optional
stocking_date	timestamp	wall-clock date/time the batch was stocked. Calendar-day based, NOT a 24-hour rolling clock — setup day = Day 0, next calendar day = Day 1 even if stocked late at night.	Optional
harvest_date	timestamp	Date when harvesting was performed.	Optional
initial_count	int	Number of valid readings included for initial.	Optional
current_count	int	Number of valid readings included for current.	Optional
harvest_count	int	Number of valid readings included for harvest.	Default: 0
total_mortality	int	Stores the total mortality.	Default: 0
harvest_weight_grams	decimal(10,2)	Stores the harvest weight grams.	Optional
initial_abw	decimal(8,2)	Stores the initial abw.	Optional
initial_abl	decimal(8,2)	Stores the initial abl.	Optional
final_abw	decimal(8,2)	Stores the final abw.	Optional
final_abl	decimal(8,2)	Stores the final abl.	Optional



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Field Name	Data Type	Description	Constraints / Default
days_in_culture	int	Calendar days elapsed since stocking_date (date-only, midnight rollover). Not a 24-hour rolling counter.	Default: 0
sample_count	int	Number of valid readings included for sample.	Optional
initial_total_weight	decimal(10,2)	Stores the initial total weight.	Optional
initial_total_length	decimal(10,2)	Stores the initial total length.	Optional
created_at	timestamp	Date and time when the record was created.	Optional

Table: sampling_records

Field Name	Data Type	Description	Constraints / Default
id	varchar	Reserved id "baseline" for the initial measurement; weekly samples use auto-id.	Primary Key
batch_id	varchar	Unique identifier of the related production batch.	Foreign Key: batches.batch_id
sampling_date	timestamp	Date and time when sampling was performed.	Optional
avg_body_weight	decimal(8,2)	Stores the avg body weight.	Optional
avg_body_length	decimal(8,2)	Stores the avg body length.	Optional
sample_size	int	Stores the sample size.	Optional
total_weight	decimal(10,2)	Stores the total weight.	Optional
total_length	decimal(10,2)	Stores the total length.	Optional
biomass	decimal(12,2)	Stores the biomass.	Optional
live_count	int	Number of valid readings included for live.	Optional
is_baseline	boolean	TRUE = initial measurement at stocking (Day 0); excluded from the weekly cadence and from Week N counting. Weekly samples are is_baseline = FALSE and are due 7 calendar days after stocking_date, then every 7 days from the previous weekly record.	Default: false
created_at	timestamp	Date and time when the record was created.	Optional

Table: mortality_records

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
batch_id	varchar	Unique identifier of the related production batch.	Foreign Key: batches.batch_id
mortality_date	timestamp	Date when the mortality was discovered or recorded.	Optional
mortality_count	int	Number of valid readings included for mortality.	Optional
created_at	timestamp	Date and time when the record was created.	Optional



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Table: harvest_records

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
batch_id	varchar	Unique identifier of the related production batch.	Foreign Key: batches.batch_id
harvest_date	timestamp	Date when harvesting was performed.	Optional
harvest_count	int	Number of valid readings included for harvest.	Optional
total_weight_kg	decimal(10,3)	Stores the total weight kg.	Optional
abw_grams	decimal(8,2)	Stores the abw grams.	Optional
created_at	timestamp	Date and time when the record was created.	Optional

Table: notifications

Field Name	Data Type	Description	Constraints / Default
id	varchar	Stores the id.	Primary Key
uid	varchar	Unique identifier of the user account.	Foreign Key: users.uid
notif_type	varchar	critical warning reminder device_auto general.	Optional
title	varchar	Short title shown to the user.	Optional
body	text	Complete message shown to the user.	Optional
is_read	boolean	Shows whether the user has opened or marked the notification as read.	Default: false
created_at	timestamp	Date and time when the record was created.	Optional